Common equations in pharmacy calculation

Calculator	Equation
Absolute bioavailability	$F = \frac{AUC_{oral} \cdot Dose_{IV}}{AUC_{IV} \cdot Dose_{oral}} \times 100\%$
First-order concentration	$C_t = C_0 e^{-kt}$
Two-compartment model	$C_t = Ae^{-\alpha t} + Be^{-\beta t}$
Steady-state concentration	$C_{ss} = \frac{F \cdot Dose}{CL \cdot \tau}$
Renal dose adjustment	$\text{Adjusted dose} = \text{Normal dose} \cdot \left(\frac{CrCl_{patient}}{CrCl_{normal}}\right)^{fe}$
Hepatic extraction ratio	$E = \frac{CL_h}{Q}$, and $F = 1 - E$
Michaelis-Menten / phenytoin	$\text{Dose rate} = \frac{V_{max} \cdot Css}{K_m + Css}$
Free drug fraction	$f_u = \frac{\text{free}}{\text{total}} \times 100\%$
Free drug amount	$\text{Free} = \text{Total} \cdot \frac{f_u}{100}$
Hill equation	$E = \frac{E_{max} \cdot C^n}{EC_{50}^n + C^n}$
Serum osmolarity	$\text{Osm} = 2Na + \frac{BUN}{2.8} + \frac{Glucose}{18}$
Osmol gap	$\text{Gap} = \text{Measured Osm} - \text{Calculated Osm}$
IV iron deficit (Ganzoni)	$\text{Deficit} = Weight \cdot (TargetHb - ActualHb) \cdot 2.4 + IronStores$
Clark's rule	$\text{Child dose} = \frac{Weight_{lb}}{150} \cdot \text{Adult dose}$
Young's rule	$\text{Child dose} = \frac{Age}{Age + 12} \cdot \text{Adult dose}$
Fried's rule	$\text{Child dose} = \frac{Age_{months}}{150} \cdot \text{Adult dose}$
Dilution / C1V1=C2V2	$V_1 = \frac{C_2 V_2}{C_1}$, Diluent $= V_2 - V_1$
Creatinine clearance	$CrCl = \frac{(140 - Age) \cdot Weight}{72 \cdot SCr}$ and $\times 0.85$ for females
eGFR CKD-EPI 2021	$\text{eGFR} = 142 \cdot \min(\frac{SCr}{\kappa}, 1)^{a} \cdot \max(\frac{SCr}{\kappa}, 1)^{-1.200} \cdot 0.9938^{Age} \cdot (1.012 \text{ if female})$
BMI	$BMI = \frac{Weight}{Height^2}$
IBW (Devine)	Male: $50 + 2.3(\text{inches over } 60)$; Female: $45.5 + 2.3(\text{inches over } 60)$

AdjBW	$AdjBW = IBW + 0.4(ActualBW - IBW)$
Vd	$V_d = \frac{Dose}{C_0}$
Half-life	$t_{1/2} = \frac{0.693}{k}$ or $t_{1/2} = 0.693 \cdot \frac{V_d}{CL}$
Loading dose	$LD = \frac{V_d \cdot C_{target}}{F}$
Maintenance dose	$MD = \frac{CL \cdot Css \cdot \tau}{F}$
Vancomycin dosing calculator	$Dose = \text{mg/kg} \cdot Weight$, with $CL_{vanc} = 0.695 \cdot CrCl + 0.05$
Phenytoin albumin correction	Corrected = Measured / $(0.2 \cdot Albumin + 0.1)$ or $(0.1 \cdot Albumin + 0.1)$ if severe renal failure
Gentamicin/Hartford dosing	Dosing weight logic uses IBW/AdjBW selection; doses are 7 mg/kg for gentamicin/tobramycin and 15 mg/kg for amikacin
Warfarin adjustment	Weekly dose is adjusted by a percentage based on INR band
Heparin infusion	Bolus and infusion rate are weight-based, e.g. 80 units/kg and 18 units/kg/hr for VTE/PE protocols
AKI staging	KDIGO criteria based on SCr ratio/rise and urine output
Corrected calcium	$Ca_{corr} = Ca + 0.8(4.0 - Albumin)$
Insulin correction	$\text{Correction} = \frac{BG - Target}{ISF}$
Insulin carbohydrate ratio	$ICR = \frac{500}{TDD}$ for rapid-acting or $450/TDD$ for regular insulin
Total bolus insulin	$\text{Total} = \text{Meal dose} + \text{Correction dose}$
Bioavailable oral dose	$\text{Oral dose} = \frac{\text{Systemic dose}}{F}$
Steady-state fraction reached	$\text{Fraction} = 1 - e^{-kt}$
Tonicity / isotonicity	NaCl required based on E-value and isotonic NaCl equivalent; 0.9% NaCl is used as the isotonic reference